

LOUGH FEA, United Kingdom

WMO: 039110

Lat: 54.7211N

Lon: 6.8147W

Elev: 227

StdP: 98.63

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.7	-5.1	2.5	-1.5	-3.7	2.8	0.0	16.8	4.7	14.2	4.9	1.5	330	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	21.4	16.6	19.7	15.7	18.2	14.9	17.5	20.1	16.6	18.6	15.7	17.4	3.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.1	18.4	15.7	11.5	17.4	14.9	10.9	16.5	50.1	20.1	47.3	18.7	44.7	17.4	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.5	9.8	8.5	DB	-5.7	23.8	2.3	1.9	-7.3	25.2	-8.7	26.3	-9.9	27.3	-11.6	28.6	(4)
(5)			WB	-6.3	19.2	2.4	1.1	-8.1	20.0	-9.5	20.6	-10.9	21.3	-12.7	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	3.6	3.9	5.0	6.9	9.7	12.3	13.9	13.7	11.8	9.0	6.0	4.0
	DBStd	4.51	2.68	2.55	2.58	2.50	2.71	2.44	2.24	2.00	2.09	2.43	2.57	3.00
	HDD10.0	1029	198	172	157	97	38	6	0	0	7	47	123	185
	HDD18.3	3651	456	406	414	342	268	182	139	145	196	289	371	443
	CDD10.0	424	0	0	1	5	29	74	121	115	61	17	2	0
	CDD18.3	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	14	0	0	0	0	0	4	7	3	0	0	0	0
	CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	4.0	4.5	4.4	4.2	3.9	3.8	3.6	3.6	3.6	3.7	4.0	4.2	4.3
Precipitation	PrecAvg	1154	117	90	82	77	83	81	82	97	87	117	121	121
	PrecMax	1419	182	175	144	122	143	166	116	200	183	213	228	244
	PrecMin	891	31	31	30	38	20	36	44	24	14	37	50	58
	PrecStd	133	41	38	27	26	34	34	27	41	46	51	46	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	11.4	13.6	16.8	21.2	22.8	24.4	23.2	19.6	15.6	13.2	11.9
		MCWB	10.3	10.2	9.8	12.2	15.5	16.3	18.7	18.5	16.5	14.2	12.6	11.5
	2%	DB	9.7	10.1	11.1	14.4	18.6	20.3	21.7	20.2	17.7	14.2	11.9	10.6
		MCWB	9.3	9.4	9.2	10.5	13.8	15.6	17.0	16.5	15.3	13.3	11.4	10.1
	5%	DB	8.8	9.0	10.0	12.5	16.3	18.5	19.6	18.4	16.4	13.4	10.8	9.6
		MCWB	8.3	8.4	8.5	9.6	12.4	14.4	16.0	15.5	14.5	12.6	10.3	9.1
	10%	DB	7.8	8.0	9.0	11.1	14.4	16.7	17.8	17.1	15.2	12.6	9.9	8.4
		MCWB	7.3	7.3	7.8	8.9	11.3	13.6	15.0	14.9	13.6	11.8	9.4	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.7	10.9	12.8	16.2	17.7	19.2	18.7	17.5	14.8	12.7	11.5
		MCDB	10.7	11.1	12.2	15.1	20.4	21.4	23.0	22.3	18.6	15.2	13.0	11.7
	2%	WB	9.4	9.5	9.8	11.4	14.4	16.2	17.7	17.3	16.0	13.7	11.5	10.2
		MCDB	9.6	9.9	10.6	13.3	17.6	19.1	20.7	19.1	17.0	14.0	11.8	10.5
	5%	WB	8.5	8.5	8.9	10.2	13.0	15.1	16.6	16.3	14.9	12.9	10.5	9.3
		MCDB	8.8	9.0	9.7	11.9	15.6	17.5	18.8	17.7	15.9	13.3	10.8	9.6
	10%	WB	7.5	7.4	8.0	9.2	11.8	14.1	15.6	15.4	14.0	12.0	9.6	8.1
		MCDB	7.8	8.0	8.8	10.7	14.0	16.3	17.4	16.6	14.9	12.5	9.9	8.4
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.9	5.8	6.6	7.3	6.9	6.5	6.3	6.0	5.2	4.6	4.4
		MCDBR	5.2	5.3	7.0	8.8	10.2	9.9	9.3	8.2	7.5	5.6	4.9	5.4
		MCWBR	5.2	5.0	5.1	5.7	6.1	5.8	5.4	4.9	5.2	4.6	4.9	5.4
	5% WB	MCDBR	5.2	5.1	6.1	7.8	9.0	8.6	8.3	7.1	6.5	5.1	4.9	5.4
		MCWBR	5.3	5.0	5.1	5.5	5.8	5.6	5.4	5.0	5.2	4.6	5.0	5.5
Clear-Sky Solar Irradiance	taub	0.333	0.343	0.366	0.378	0.383	0.373	0.375	0.359	0.353	0.340	0.326	0.321	
	taud	2.204	2.286	2.263	2.270	2.297	2.359	2.357	2.417	2.430	2.395	2.267	2.171	
	Ebn at Noon	576	728	798	845	860	873	863	858	812	728	586	495	
	Edh at Noon	63	81	103	117	121	115	113	100	87	72	58	52	
All-Sky Solar Radiation	RadAvg	0.57	1.25	2.28	3.70	4.70	4.79	4.39	3.57	2.57	1.47	0.70	0.42	
	RadStd	0.06	0.11	0.22	0.35	0.43	0.48	0.36	0.30	0.13	0.13	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page